

LAB 3 • ANALYSIS AND COMMUNICATION

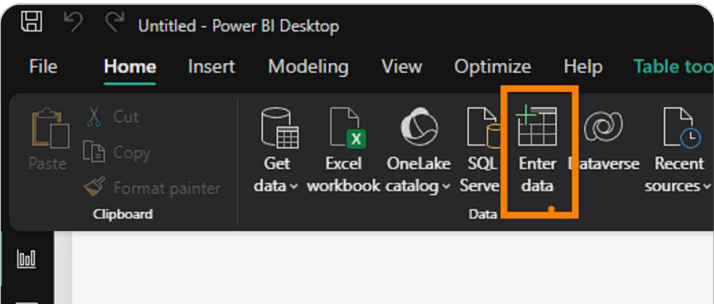
Build a report that supports decisions

Create reusable DAX measures, then organize visuals around three questions: Where are we over budget? Where is procurement exposure concentrated? What should leadership review first?

35–45 minutes Report view + DAX Deliverable: 3 report pages

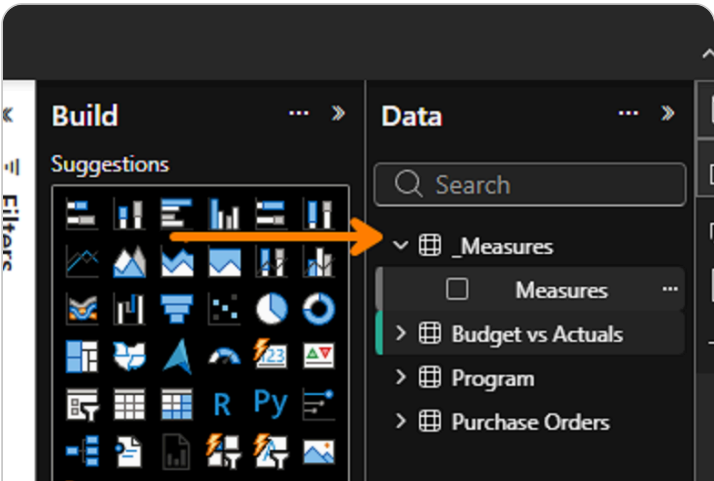
Create a measures table

- 1. Select **Home** → **Enter data**.
- 2. Name the table **_Measures**, name its single column **Measures**, enter one value of 1, and select **Load**.
- 3. Hide the Measures column in Report view after you have created the measures.
- 4. For each formula below, select **New measure** and set its Home table to **_Measures**.



Start with Enter data

Create a small placeholder table so all report measures have one predictable home.



Keep measures together

The `_Measures` table appears at the top of the Data pane and becomes the organized collection for the seven measures below.

Why measures? Measures calculate in the current filter context. The same formula works correctly in a card, chart, matrix, drillthrough page, or tooltip.

Core measures

Total Budget = SUM('Budget vs Actuals'[Budgeted Amount])

Total Actual = SUM('Budget vs Actuals'[Actual Amount])

Budget Variance = [Total Budget] - [Total Actual]

Budget Variance % = DIVIDE([Budget Variance], [Total Budget])

Total PO Value = SUM('Purchase Orders'[PO Amount])

Outstanding PO = SUM('Purchase Orders'[Amount Remaining])

PO Count = DISTINCTCOUNT('Purchase Orders'[PO ID])

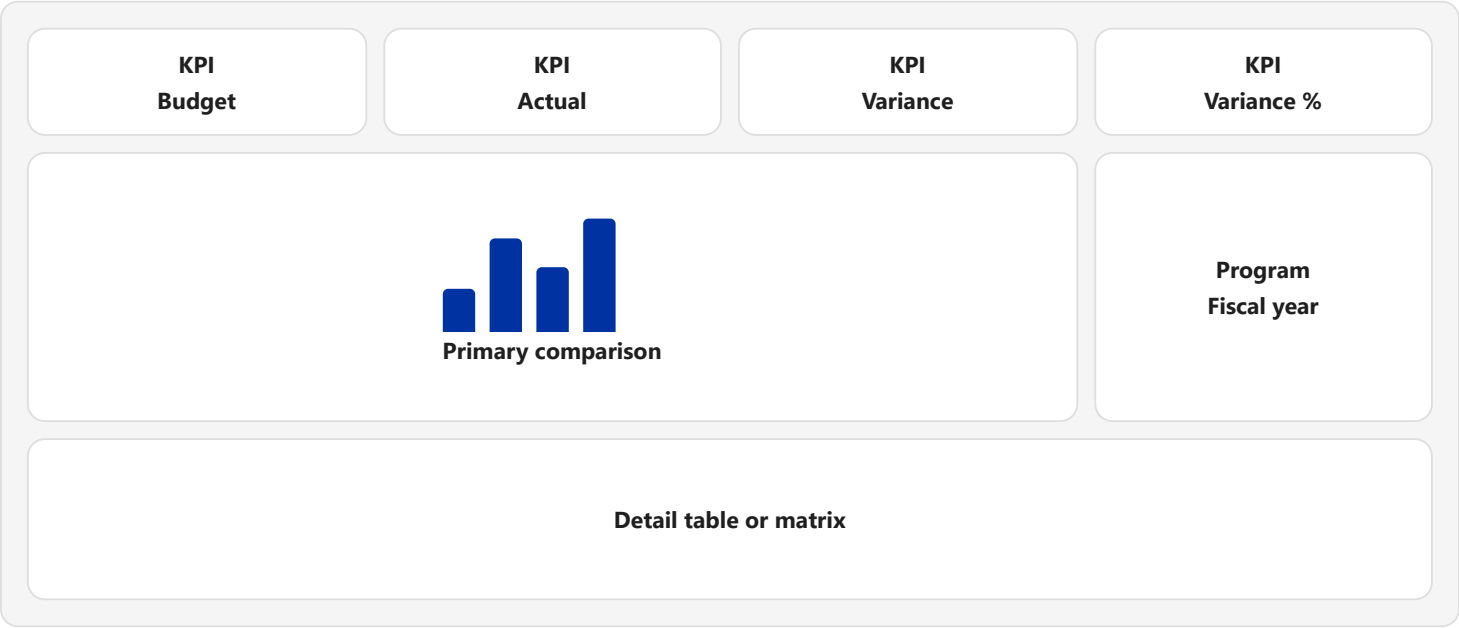
Format money measures as Currency with 0 decimal places. Format Budget Variance % as Percentage with 1 decimal place.

Match the completed-report presentation

1. Set each page to a custom **1920 × 1080** canvas (16:9).
2. Add a clear page heading: Financial Health, Procurement Exposure, or Executive Review.
3. Use a light neutral canvas, white visual surfaces, subtle gray borders, and a restrained navy/teal finance palette.
4. Use explicit chart and table titles that state the decision being supported.
5. Keep 24–32 px gaps between visuals and align edges across each row.
6. On Executive Review, add two **Page navigation** buttons targeting Financial Health and Procurement Exposure. Keep both controls in the top-right header band.

Completed-report design choice: Financial Health uses four individual KPI cards. Procurement Exposure and Executive Review each use one grouped card visual containing three measures, which keeps related KPIs aligned and reduces container clutter.

Page layout pattern



Keep the reading order consistent: headline KPIs, primary comparison, filters, then detail.

Build three focused pages

Page 1 · Financial Health

Question: Which programs and periods are driving the budget gap?

Individual cards	Total Budget, Total Actual, Budget Variance, Budget Variance %
Bar chart	Axis: Program[Program Name]; Values: Total Budget and Total Actual
Line chart	Axis: Budget vs Actuals[Fiscal Period]; Values: Total Actual; Legend: Program[Program Name]
Matrix	Rows: Program Name; Columns: Fiscal Period; Values: Budget Variance
Slicers	Program[Branch], Fiscal Year, Budget Category

Apply conditional formatting to Budget Variance: negative values need attention. Show data labels in millions.

Page 2 · Procurement Exposure

Question: Which vendors and orders hold the largest unpaid commitments?

Grouped KPI card	Total PO Value, Outstanding PO, PO Count
Bar chart	Axis: Vendor Name; Value: Outstanding PO; sort descending; Top N = 10
Column chart	Axis: Program[Program Name]; Value: Total PO Value
Donut	Legend: PO Status; Value: PO Count
Table	PO ID, Vendor Name, Delivery Date, PO Amount, Amount Remaining, Priority
Slicers	Program[Program Name], Priority, PO Status

Conditionally format Priority and Amount Remaining. Keep the numeric value visible so color is not the only signal.

Page 3 · Executive Review

Question: What requires leadership attention now?

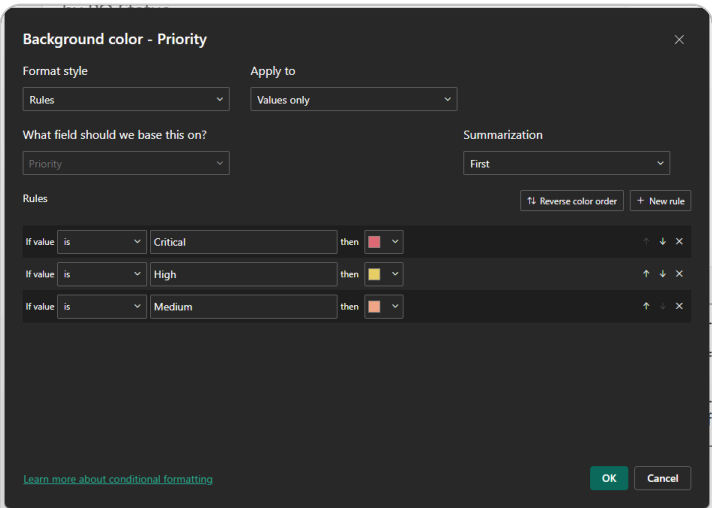
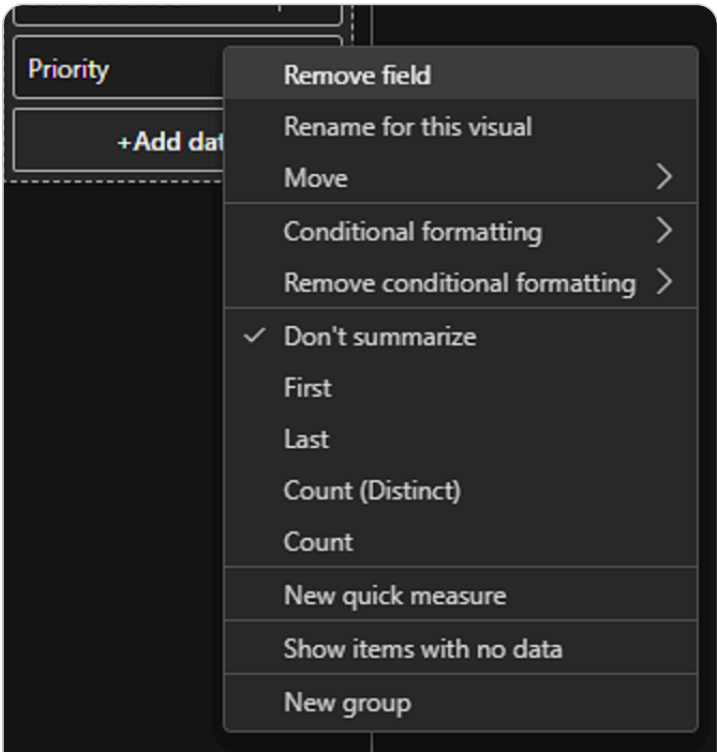
Grouped KPI card	Budget Variance, Budget Variance %, Outstanding PO
Bar chart	Program Name with Budget Variance, sorted ascending so the worst gap appears first
Matrix	Program Name, Total Budget, Total Actual, Budget Variance, Outstanding PO
Slicers	Branch and Fiscal Year
Navigation	Page-navigation buttons to Financial Health and Procurement Exposure

Keep this page sparse enough to scan during a meeting and place navigation consistently in its header.

Apply conditional background formatting

On the Procurement Exposure table, use color to make Priority easier to scan while keeping the text visible.

1. Select the table visual.
2. In the Build pane, open the menu beside **Priority**.
3. Select **Conditional formatting** → **Background color**.
4. Set **Format style** to **Rules**, **Apply to** to **Values only**, and base the rules on **Priority** summarized by **First**.
5. Add text rules for **Critical** (red), **High** (yellow), **Medium** (peach), and **Low** (green).
6. Select OK, then confirm every row still shows its Priority text over the background color.



Define text-based rules

Use Rules and Values only, then assign a distinct accessible color to each Priority label. Add a fourth rule for Low.

Do not rely on color alone: retain the Priority text and use colors with enough contrast. Apply numeric data bars or a color scale separately to Amount Remaining if useful.

Configure interactions

1. Select a chart, then **Format** → **Edit interactions**.
2. Let Program and Branch selections filter all relevant visuals.
3. Use highlight only when the retained total provides useful context.
4. Keep cards filtered; leadership should see that a slicer changed the scope.
5. Test one program selection on every page, then clear all filters.

Accessibility and finish

- Use descriptive titles such as "Outstanding PO value by vendor," not "Bar chart."
- Add alt text under **Format** → **General** → **Properties** for each meaningful visual.
- Do not communicate status through color alone; retain labels and values.
- Use consistent currency display units and decimal precision.
- Align visuals and keep page navigation in the same location.

Validate the unfiltered report

\$234.72M Total Budget	\$265.20M Total Actual	\$23.855M Total PO Value	\$16.095M Outstanding PO
--	--	--	--

The budget variance should be **-\$30.48M**, approximately **-13.0%**. A mismatch usually indicates incorrect percentage/currency conversion, an unremoved duplicate, or a filter that remains selected.

Final test

1. Select one Program Name and confirm both budget and procurement visuals filter.
2. Select FY2026 and confirm the procurement page does not change unless a date relationship/filter was intentionally added.
3. Clear all filters and compare the totals above.
4. Save the PBIX, close it, reopen it, and select **Refresh**.

Important modeling observation: Fiscal Year belongs only to Budget vs Actuals in this dataset. Do not imply that it filters purchase orders unless you add a proper shared Date dimension.